Track 2 — iteration 5: start from the alleles, not from the drug databases

Standing: draft, 2026-09-25, from iteration 5 step 1 only (entries 158–162). **Not uploaded.** Track 2 accepts one submission per team; the user plans to merge the Track 2 iterations into one upload, and this folder is an input to that.

Case. One child with mosaic variegated aneuploidy (MVA), carrying two different broken copies of `BUB1B` — a *compound heterozygote*: one `stop_gained` allele (a premature stop signal that cuts the protein short) and one missense allele (a single amino-acid change). The pair was confirmed by the organisers' oracle in Track 1 (entry 083). `BUB1B` encodes **BubR1**, a component of the spindle assembly checkpoint (the brake that holds cell division until every chromosome is correctly attached).

Two data sources, everywhere. The **genome** (the delivered variant callset and reads) fixed the gene, the two alleles, their consequences and the child's pharmacogenome. The **clinical dataset** (the proband's 8 HPO terms) disqualified drug classes on this child's own phenotype axes and selected the mouse that matches him. Every section below says which of the two it used.

This is a research exercise. Nothing here is advice for or against giving any drug to any patient, and no number here is a probability of benefit.

---

1. How this report answers the panel's four questions

criterion	weight	where	what it rests on	the weakest point, stated by us
Scientific rigor	35 %	§5–§8, §12	four routes, **each with a positive control declared in writing before the child's data were touched**; two of those controls **failed and are reported as failed**; every table carries a `.provenance.json` with input sha256s	this iteration files **no pre-registered hypotheses** (§3). A route that passes its control is *a validated method applied once*, not a tested claim
Potential impact	25 %	§9	a named next measurement worth more than any drug choice on the table; an **avoid list** built from the child's own genome, his own HPO terms, and what MVA patients were actually given	the highest-ranked drug is a weak, mechanism-free hit (§6); the impact is in the *test*, not the prescription
Innovation	25 %	§10	readthrough priced **per allele** against a dosage arithmetic; **proliferation-corrected** protein-level drug screening for a cell-cycle gene; a **code-enforced quote gate** over model-read literature; a decision layer that reports **EVPI/EVSI** instead of a ranked list	the decision layer is a transparent heuristic, not a calibrated posterior
Scalability	15 %	§11	all three discovery routes re-run on **Bloom syndrome** from case cards; 587 configuration lines against entry 156's 378-line purpose-built driver; every first-case table byte-identical	"configuration only" was **false** as the routes stood: 271 lines of parameterisation were needed first, and we report that number, the 357 that includes the forced module splits, and the 491-line full diff

---

2. The answer in one page

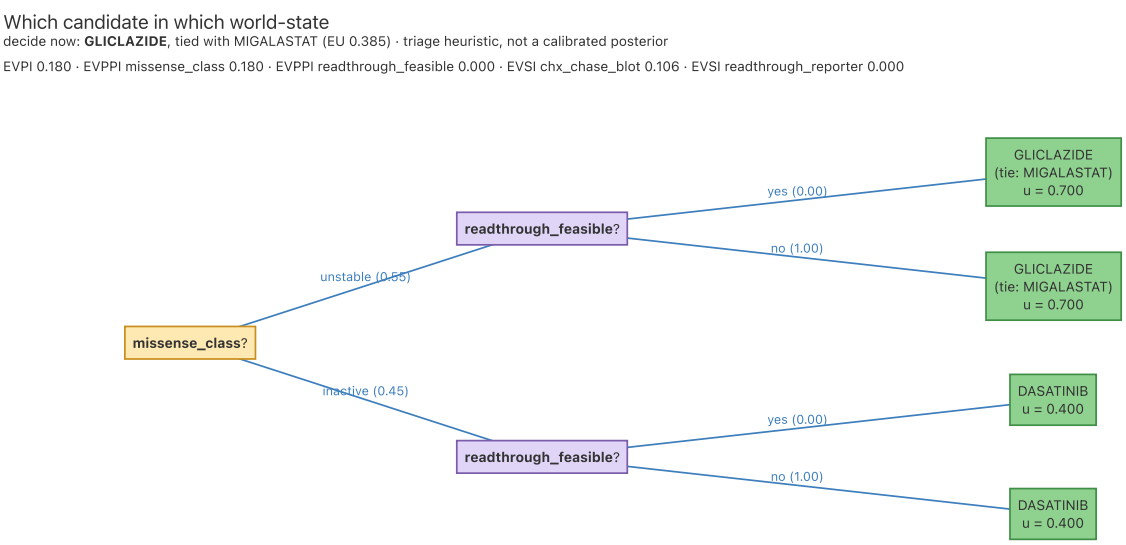
> **Rubric:** all four. The short version, with every number linked to the section that measured it.

The question this iteration asked. Iterations 2 and 4 asked drug databases what to give this child, and the databases answered at chance level (§3). Iteration 5 asks a different question: *given these two specific alleles and this protein, what could raise the amount of working BubR1, and how much would each route actually be worth?*

Four findings.

1. **The stop allele is not the lever.** It is NMD-competent (its message is destroyed by the cell's quality-control system) and it deletes the entire kinase domain. A readthrough drug — one that makes the ribosome read past a premature stop — is worth **at most +1.4 percentage points** of working BubR1 against a gap of about **6.5 points**, even with every assumption at its most generous. Doubling the missense protein's abundance is worth **+12.5 points** on the same arithmetic (§5).
2. **The amount-raising branch is nearly empty of approved drugs.** With proliferation removed (BubR1 rises and falls with the cell cycle, so uncorrected readouts measure growth, not BubR1), only 18 of 875 compounds raise BUB1B protein; the 5 approved ones raise it by **+14–16 %**. The strongest raisers are a CDK1 inhibitor (+88 %) and a proteasome inhibitor (+45 %) — unapproved and non-specific, but they name the lever: **degradation** (§6).
3. **The literature holds what no database encodes**, and every row is machine-verified against its source text. 30 intervention rows and 12 patient-exposure rows kept, 0 rejected, gate mutation-tested; NAD^+/NMN, HDAC inhibitors, proteasome inhibitors and senolytics. It also yields something no drug database has: **what MVA patients were actually given** — a severe dactinomycin reaction in a biallelic-`BUB1B` case, published advice to avoid vincristine or taxol, growth hormone ineffective across four reports (§7).
4. **The decision is dominated by one unmeasured fact, and we can price it.** Everything turns on whether the missense protein is *unstable* (too little of it) or *inactive* (enough, but broken). **EVPI is 0.180, all of it on that one variable**; the readthrough question is worth **0**. A cycloheximide-chase Western blot (a bench test that measures how fast the protein decays) captures **0.106**, about **59 %** of that ceiling. The best action flips at P(unstable) = **0.414**, and our prior sits at 0.55 — only 0.14 above the flip (§8).

What we would say to a treating team. Not "give drug X". The ranked list's top entry (gliclazide, a diabetes drug, tied with migalastat) rests on a 14–16 % protein rise in one cancer cell line with no mechanism argument, and we say so rather than dressing it up. The defensible output is the **avoid list** (§9.2) and the **one measurement** that would change what anyone should do next (§9.1).



> **Legend** yellow = the first uncertain fact · purple = further facts about the child's alleles · each edge = a value with its prior weight · green = the best drug in that state · red = a state in which only the baseline (defer, run the test first) is best.

---

3. Why this iteration is shaped differently — and what it gives up

> **Rubric:** Rigor and Innovation. A method change has to be argued from the previous method's failures.

What iterations 2 and 4 measured.

earlier route	outcome	the lesson taken
PrimeKG knowledge-graph walk (G1, entry 142)	vitamins and metals rank top	the walk scores **connectivity**, not mechanism
screen-wide PRISM sweep (G3, entry 154)	nothing survives multiple-testing correction (best q 0.122)	cancer cell lines are a poor proxy for a constitutional hypomorph
LINCS signature reversal (G4, entry 153)	**positive control failed**	an mRNA signature of a cell-cycle gene reads as generic arrest
convergence of all four (H094, entry 155)	one shared drug, p 0.263 = chance	weak independent sources do not add up
the phenotype twin (entry 150)	NAD⁺/SIRT2 and senolysis, with real rescue data	**the mechanism-first route is the one that found something**

The design that follows. Start at the two alleles and the protein; let each route be a *measurement* about this genotype rather than a vote from a database; and make every route prove itself on a case it cannot get wrong before it is pointed at the child.

What this iteration gives up, explicitly. It files **no hypotheses** in the science register (the user's decision, recorded in `notes/exploratory-no-hypotheses.md`). Instead, **each route declares a positive control in writing before it runs**. That is weaker than pre-registration in one specific way — the analysis could have

been steered after seeing the data — and stronger in another: a route whose control fails is reported as uninterpretable rather than argued around. **Two of the six controls in this report failed** (§6.3, §11), which is the only evidence we can offer that the discipline was applied rather than announced.

---

4. The genotype and the mechanism it implies

> **Rubric:** Rigor. What the two data sources fixed before any drug was considered.

fact	value	source
gene	`BUB1B`, transcript `ENST00000287598` (Ensembl 116), 23 exons, 1,050 residues	genome
allele A	`stop_gained`, **UGA**, exon **17 of 23**, **746 nt** upstream of the last exon–exon junction with 6 junctions after it ⇒ **NMD-competent**	genome, entry 158
allele B	missense, predicted destabilising; ΔΔG cannot separate published allele classes at n = 8, so the class is **predicted, not measured**	genome, entry 149
protein lost by allele A	**736 of 1,050 residues kept (29.9 % lost)**; the **entire protein-kinase domain** (UniProt 766–1050) is gone	entry 158
mechanism class	**loss of function, hypomorphic** (some working protein, too little of it) — four of five independent instruments agree	iteration 4 §3
the matching mouse	the BubR1 **hypomorph** (H/H, H/L1002P), selected from all 76,699 MGI genotypes by the child's 8 HPO terms	clinical, entry 150

Why the mouse matters here. The hypomorph is the genotype in which NAD^+/SIRT2 and senescent-cell clearance were shown to work. A knockout mouse would have answered a different question. That selection came from the **clinical** dataset alone, independent of the genome.

---

5. Route R1 — can the stop allele be read through, and is it worth reading through?

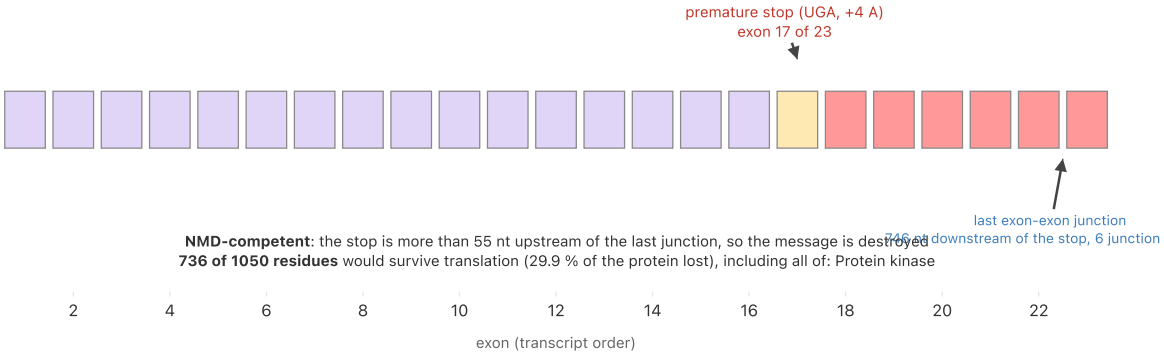
> **Rubric:** Rigor and Innovation. The first route prices a therapy class against this child's exact allele rather than against his gene. **Entry:** 158. **Sources:** genome (the allele), clinical (which drug classes are disqualified).

The idea. Some drugs make the ribosome insert an amino acid at a premature stop instead of halting — *nonsense readthrough*. How well that works is not a property of the gene but of the **stop codon and the few bases around it**, and it competes with NMD, which destroys the message before it can be translated at all.

5.1 The allele's anatomy

fact	value	why it matters
stop codon	**UGA**	the most readthrough-permissive of the three stops, and the only one DAP acts on
+4 base (immediately after the stop)	**A**	the single strongest context feature; `C` is best, `A` is second for DAP and near-worst for aminoglycosides
–1 base	**G**	a weaker modifier
NMD	**competent** (746 nt upstream of the last junction; the 50–55 nt rule)	the message is destroyed, so there is little substrate to read through
domains lost	the **whole kinase domain**	the domain the child's *other* allele also sits in

The stop allele's anatomy: exon, decay verdict, and what is lost



> **Legend** purple = coding exons before the stop · yellow = the exon carrying the stop (the decision point for the 50–55 nt rule) · red = the exons the truncated protein never reaches · blue text = the junction the rule is measured against.

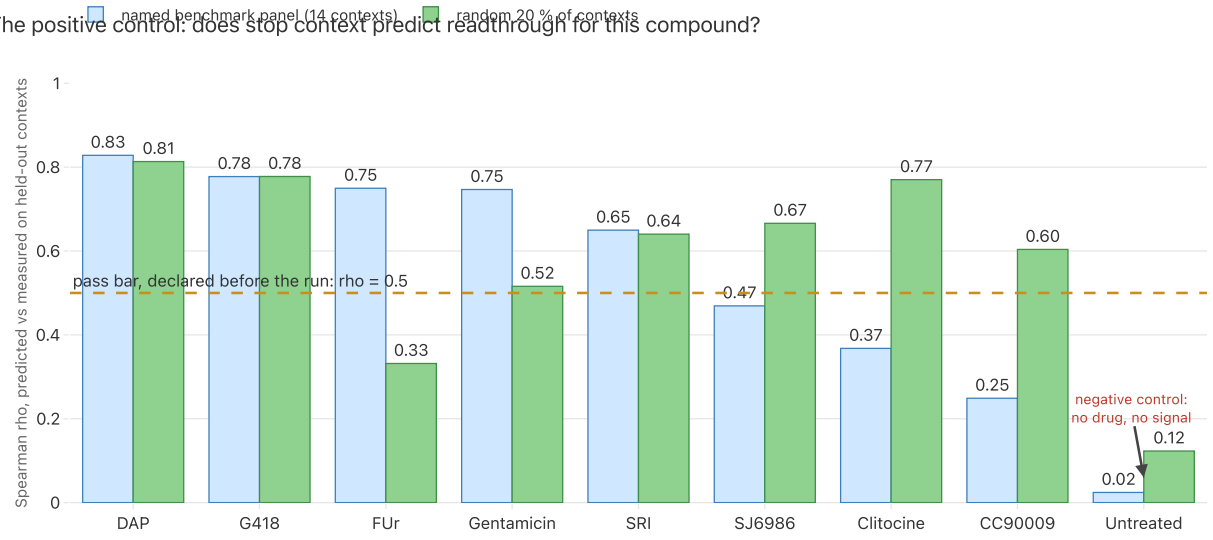
5.2 The positive control, declared before any prediction

The model is a context model fitted to **Toledano, Supek & Lehner, *Nat Genet* 2024 (PMID 39174735)** — 5,837 **measured** premature stops × 8 compounds. The child's stop is **not** in that dataset, so every number for him is a prediction. The control was two held-out sets: 14 literature benchmark stops, and a random 20 %.

compound	ρ (14 held out)	ρ (random 20 %)	verdict
DAP (2,6-diaminopurine)	**0.83**	**0.81**	**passes both**
G418	**0.78**	**0.78**	**passes both**
SRI-41315	**0.65**	**0.64**	**passes both**
Gentamicin	0.75	0.52	fails on ordering
Clitocine · SJ6986 · CC90009 · 5-fluorouridine	0.25–0.77	0.33–0.77	fail one arm each
Untreated (negative control)	**0.02**	**0.12**	**fails, as it must**

The untreated column is the load-bearing part: run through the identical code path, it orders at chance. A pipeline that "predicted" readthrough from untreated data would have been detected here.

The positive control: does stop context predict readthrough for this compound?

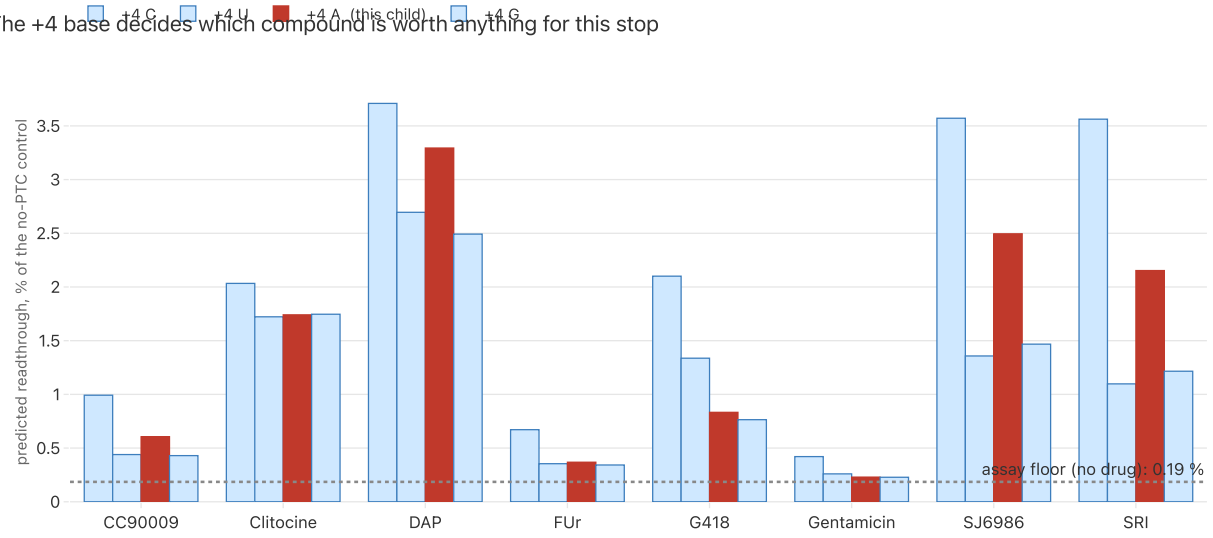


> **Legend** blue = the named benchmark panel, the declared primary control · green = the random held-out fifth · yellow dashed = the pass bar fixed before the run · the red note marks the dataset's own no-drug column, which must fail and does.

5.3 What the child's stop would do, and what it would be worth

Predicted readthrough, as a percentage of a no-stop control (no-drug floor 0.18 %): **DAP 2.85**, SRI-41315 2.34, G418 0.58, gentamicin 0.21. The **+4 A** context costs G418 a factor of 3.6 but DAP only 1.3 — an allele-level reason the aminoglycosides are the wrong class here, quite apart from the clinical one below.

The +4 base decides which compound is worth anything for this stop



> **Legend** red = the +4 base this child's stop actually has · blue = the three bases it does not · grey dotted = the assay's no-drug floor, below which a prediction is noise.

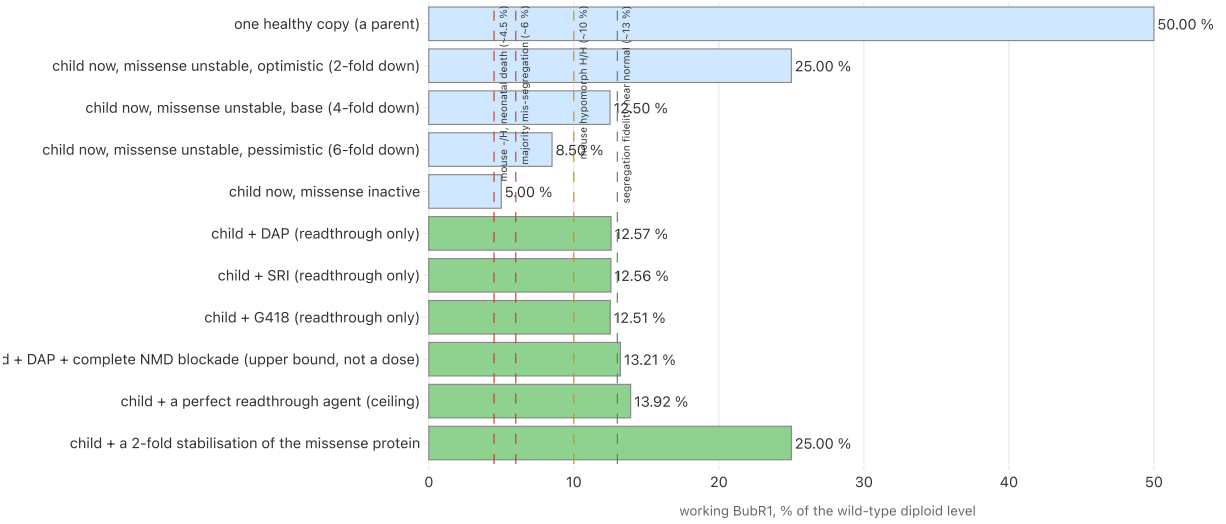
The dosage arithmetic converts that into working protein:

scenario	working BubR1 (% of wild type)	drug adds
one healthy copy (a parent)	50.00	—
child now (base case)	**12.50**	—
child + DAP	12.57	**+0.07 pt**
child + DAP **and a complete NMD blockade** (a ceiling, not a dose)	13.21	+0.71 pt
a perfect readthrough agent (full transcript rescued, best measured readthrough, wild-type product)	13.92	**+1.42 pt**
a 2-fold stabilisation of the missense protein	**25.00**	**+12.5 pt**

External check: the base case lands between the mouse hypomorph H/H (~10 %) and the ~13 % level at which human cells segregate chromosomes near-normally (PMID 20516114) — which is where a viable MVA child should land.

Conclusion. Closing the gap needs about 6.5 points, i.e. ~25 % readthrough; no compound in the dataset exceeds about 5 % on any context. **The stop allele is not the lever. The missense allele is.** That conclusion is what makes §6 and §8 the centre of this report.

What each lever is worth: readthrough moves the total by fractions of a point



> **Legend** blue = untreated states · green = scenarios with a treatment · dashed = published genotypes as yardsticks (green safe, yellow the viable hypomorph, red the failing side).

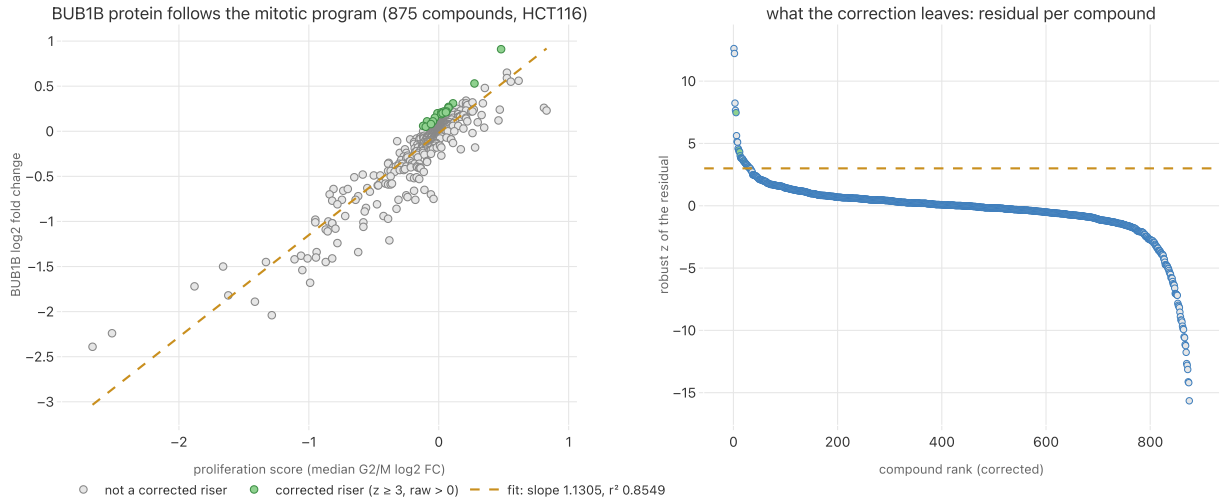
Both datasets. The genome supplied the allele and its context; the **clinical** terms disqualified the aminoglycoside class independently, because gentamicin is nephrotoxic and the child carries **nephrocalcinosis (HP:0000121)**. Ataluren scores **0**: its EU authorisation was not renewed (EC decision 28 Mar 2025), and its original signal has a documented luciferase artefact (PMID 19208811).

---

6. Route R2 — which drugs raise BubR1 *protein*, once growth is taken out?

> **Rubric:** Rigor and Innovation. Direction D3 ("raise stable BubR1") is about protein amount, so it should be measured on protein, not mRNA. **Entry:** 159. **Source:** genome (the gene); this route does not use the clinical terms, and says so.

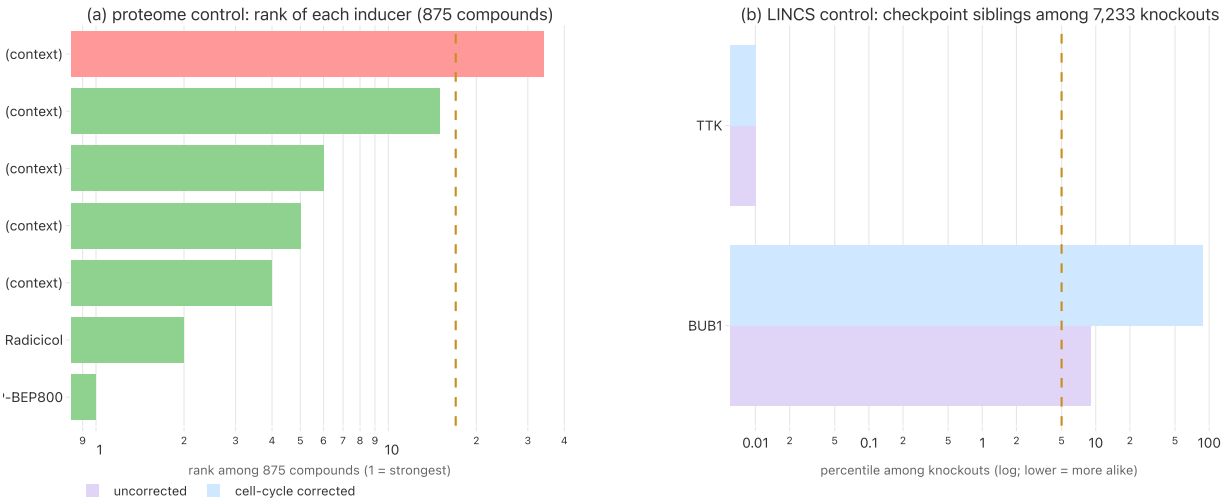
The confound, stated first. BubR1 is a mitotic protein: anything that slows or speeds division moves it. In CCLE, BUB1B protein correlates with a proliferation score at **r 0.704** across 378 cell lines; in the drug-perturbation data, **r 0.925 (r^2 0.855)**. BUB1B sits in the **99.7th percentile** of proliferation-bound proteins. Any uncorrected "BUB1B raiser" is a growth artefact. We therefore score each compound on the **residual** after regressing out a proliferation score defined in advance from a published G2/M gene set.



> **Legend** left: each dot is one compound, the dashed line the fitted proliferation dependence · right: the same compounds by corrected rank, dashed line = the z ≥ 3 cut · green = a corrected riser, grey = context, not the subject.

6.1 The positive control, declared before BUB1B was looked at

In the primary dataset (Mitchell 2023, PMID 36593396 — 9,714 proteins × 875 compounds), HSP90 inhibition must raise the heat-shock protein HSPA1A. It ranks **1st (z 46.7) and 2nd (z 39.4) of 875**, against a "top 2 %" cut: **passed**. In the deep sets, bortezomib → HSPA1A ranks 1st in all three cell lines; the nutlin → MDM2 control **failed in A549 and RKO**, so those two datasets are marked uninterpretable and the rows they contribute are capped at strength 0.10.



> **Legend** (a) the proteome control: green = the inducer ranks inside the gating cut, red = outside it · (b) the LINCS control: purple = uncorrected, blue = cell-cycle corrected; BUB3, MAD1L1 and MAD2L1 have no knockout row and are not drawn · dashed = the 5 % cut.

6.2 The result, and why it is thin

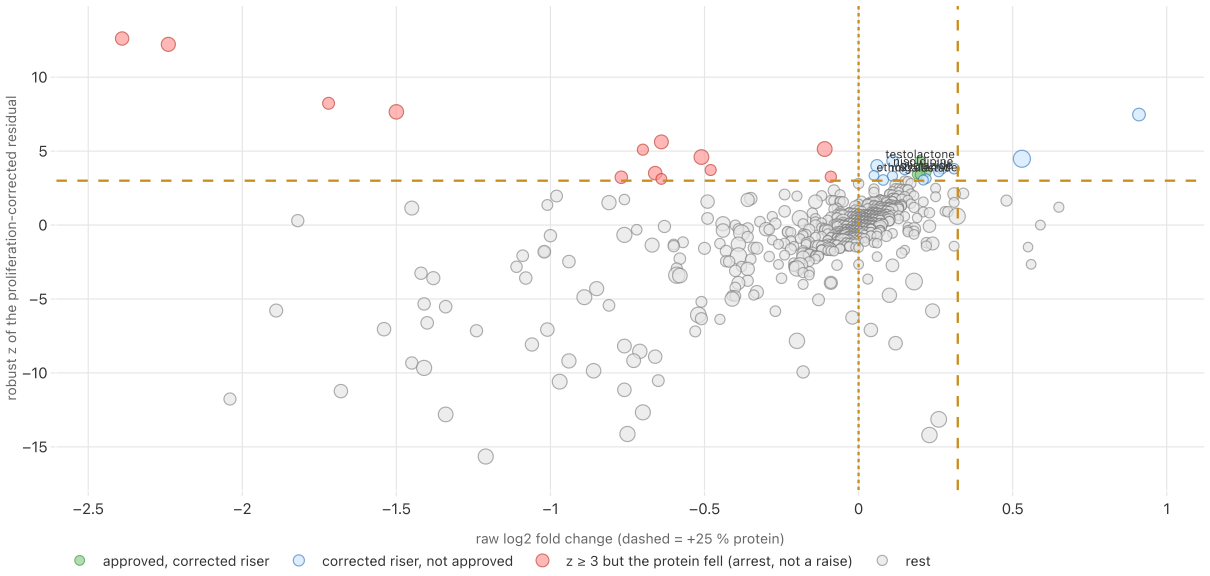
18 of 875 compounds raise BUB1B protein after correction. **Five are approved**, and all five raise it by only **+14–16 %** (log2FC ≈ 0.2, robust z 3.4–4.3):

drug	usual indication	robust z	rank of 875
testolactone	breast cancer (withdrawn in several markets)	4.33	12
nisoldipine	hypertension	3.85	14
gliclazide	type 2 diabetes	3.59	20
ethoxzolamide	glaucoma	3.44	22
migalastat	Fabry disease (a pharmacological chaperone)	3.42	23

None of these has a mechanism argument for BubR1. They are the approved tail of a correlation-corrected screen in one colorectal cancer line. The largest raisers are unapproved and non-specific — **Ro-3306** (a CDK1 inhibitor, +88 %) and **epoxomicin** (a proteasome inhibitor, +45 %) — and their value is not as candidates but as a pointer: **the lever is degradation**, which is exactly the branch §8 says is worth measuring.

Of the five, **migalastat** is the only one whose mechanism class (a chaperone that stabilises a misfolded protein) is even conceptually aligned with the missense-unstable branch. We note that as a hypothesis for a later iteration, not as evidence here: its target is an unrelated enzyme.

BUB1B protein: corrected response vs effect size (875 compounds, HCT116, 24 h)



> **Legend** green = an approved corrected riser · blue = a corrected riser that is not approved · red = z ≥ 3 but the protein *fell* (arrest, not a raise) · grey = every other compound · marker size = breadth (the share of all proteins that compound raises) · dashed horizontal = z ≥ 3, dashed vertical = +25 % protein, dotted = no change.

6.3 The second part: a corrected LINCS signature — control failed

Entry 153's mRNA route failed its control because losing `BUB1B` looked like generic arrest. We re-ran it with the proliferation component removed, then re-ran **its own** control: does the `BUB1B`-loss signature resemble the loss of the other checkpoint genes?

It does not. Mean sibling percentile 44.23, permutation **p 0.390** (the PC1/PC3/PC10 variants also fail). Reversal was therefore **not run**, and this route contributes **zero** candidate rows.

Two honest observations come out of the failure:

- The **uncorrected** baseline *passes* by the same rule (4.53 %, p 0.0041). So entry 153's failure was partly a query-formulation problem, not only biology — and our correction, which is the more defensible analysis, makes the result *worse*. We report the harder number.
- **TTK ranks 1st of 7,233 knockouts in every variant** (r 0.42 → 0.257 after correction) — the single checkpoint relationship that survives. TTK inhibitors are on the **avoid** list (§9.2), because weakening this child's checkpoint further is the one direction the mechanism forbids.

---

7. Route R3 — every intervention the literature reports, each backed by a verified quote

> **Rubric:** Rigor and Innovation. The strongest lead of iteration 4 came from a paper, not a database; this route makes that repeatable and auditable. **Entry:** 160. **Sources:** both — the genome names the gene, the clinical terms filter the exposures.

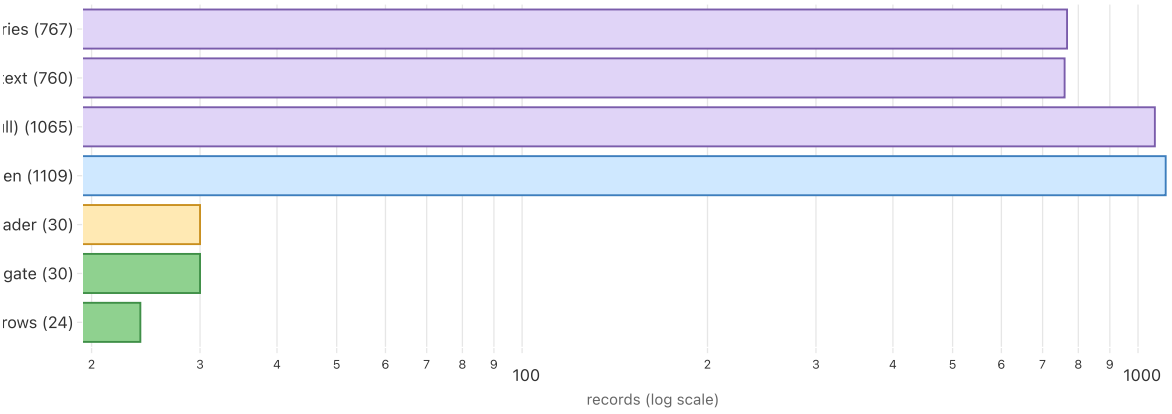
Why it exists. No database has an edge saying "compound X raises BubR1 levels". That fact lives in sentences. The risk of having a model read those sentences is fabrication, so the **reproducible gate is code, not trust**: a verifier normalises whitespace and Unicode and rejects any row whose quote is not a **verbatim substring** of the cached source text.

stage	count
Europe PMC queries (no PMID, author or title seeded)	6
papers retrieved	767 (760 with text; 1,065 cached texts, 307 open-access full texts)
sentences screened	79,435
sentences kept by the lexical screen	1,109 (1.4 %) in 283 papers
rows written	**30** intervention + **12** patient-exposure = 42
rows surviving the quote gate	**42 kept, 0 rejected**

Zero rejections is not proof that a gate works, so the gate is **mutation-tested**: flipping `increases` → `decreases` inside a true quote makes the row fail with `quote_not_in_source`, as do too-short quotes, unknown PMIDs and out-of-vocabulary terms.

Positive control, declared before the run: queries written without seeding them must recover North 2014 (NMN/SIRT2, PMID 24825348) and the two senescent-cell-clearance papers (18516091, 22048312), and produce a row for each. **Retrieval 3/3, extraction 3/3 — passed.**

R3 extraction funnel: from a query to a quote-verified candidate



> **Legend** purple = the raw corpus · blue = each machine stage · yellow = the quote gate, the decision point · green = the rows that survive it. Log x-axis, so a short bar is still hundreds.

7.1 What it found

intervention (as reported)	effect	approved class analogue	strength
NMN / NAD⁺ precursor	raises BubR1, extends life in the hypomorph	nicotinamide, niacin	0.192
MG132 (proteasome inhibition)	raises BubR1	bortezomib, carfilzomib	0.192
TSA (HDAC inhibition)	BubR1 acetylation axis	vorinostat, romidepsin, belinostat, panobinostat, valproic acid	0.144
demethylation	BubR1 expression	decitabine, azacitidine	0.144
FTI-277	BubR1 axis	lonafarnib	0.144
senescent-cell clearance	delays the hypomorph's ageing phenotypes	dasatinib (0.134), quercetin, fisetin, navitoclax (0.067)	

R3 evidence map: every quote-verified intervention × what it moved



> **Legend** green = the direction a hypomorph wants · red = the wrong direction · grey = no agreed good direction. A big square is an animal or a patient, a small one is cells.

The direction check matters more than the hits. Every approved row above is a *declared class analogue*: the gene → drug path returned **zero** mechanism molecules for SIRT2, CDKN2A, PELI1 and `BUB1B` itself. It did map `CHEK1` (9 inhibitors) and `KDM1A` (bomedemstat) to **harm** rows — knockdowns that *lower* BubR1 — and the polarity filter kept them out of the candidate list and onto the avoid list. A route that only collected "related drugs" would have proposed them.

7.2 What MVA patients were actually given — the finding no drug database holds

12 quote-verified exposure rows:

finding	PMID	relevance to this child
a biallelic- `BUB1B` Wilms tumour case had a **severe adverse reaction to dactinomycin**; chemotherapy was abandoned	31081598	**the same gene**
a `BUB1B` PCS/MVA report advises **avoiding vincristine or taxol**	28611924	vincristine is standard in rhabdomyosarcoma — this child's **HP:0002859**
the only MVA1 stem-cell transplant: reduced-intensity treosulfan tolerated, then graft rejection, sepsis, death day +156	31053147	a whole modality's record in this disease
growth hormone ineffective across 4 reports	—	against the child's short stature **HP:0004322**

What MVA patients were actually given, and how it went



> **Legend** green = tolerated · yellow = a response question · red = toxicity or death · blue = an advisory statement.

---

8. Route R4 — which branch, and what is the next measurement worth?

> **Rubric:** Innovation and Impact. A ranked list answers "what would you give?". A clinician's real question is "what should we do next?", and those have different answers here. **Entry:** 161. **Sources:** both — R1–R3's candidate rows (genome-derived) and the per-child safety table (genome + clinical).

The model, stated plainly. Two unmeasured state variables split the world:

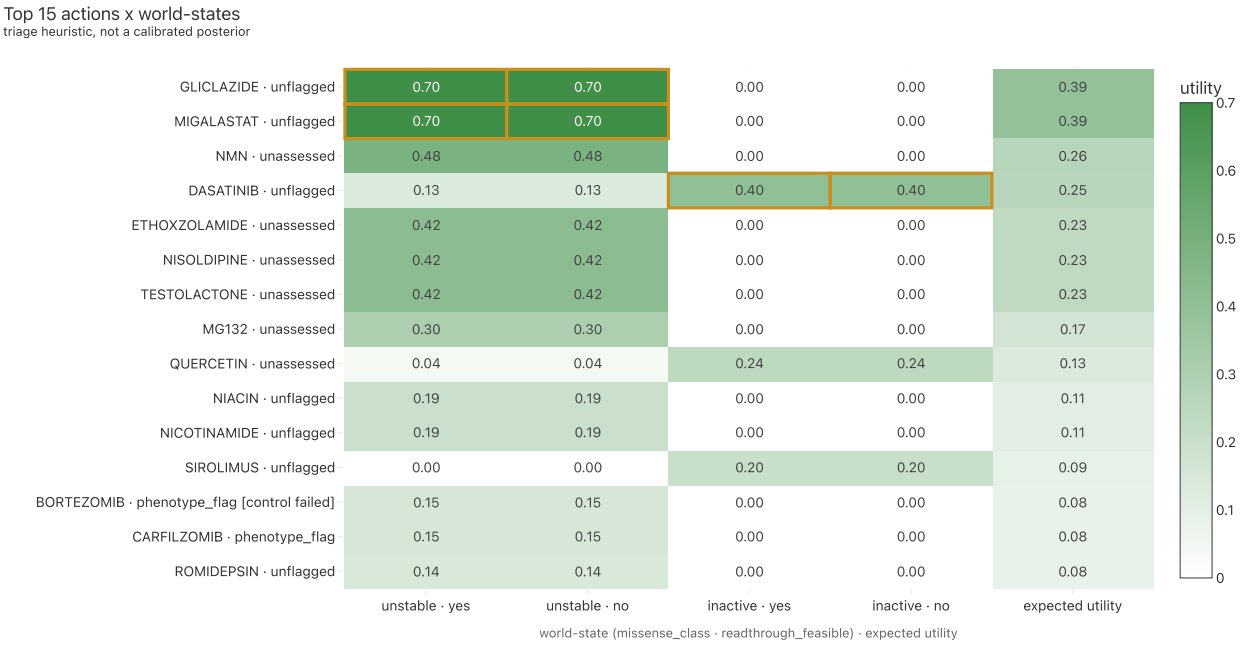
- `missense_class` — is the missense protein *unstable* or *inactive*? Prior **0.55 / 0.45**, carried over from entry 149's stability work. No table in this iteration measures it.
- `readthrough_feasible` — **derived, not typed**: "feasible" means a readthrough dose lifts an untreated scenario past R1's 13.0 % fidelity cliff. Read from `allele_dosage.tsv`, **0 of 9 real dose pairs cross it → P(yes) = 0**; with R1's two upper bounds included, 2 of 15 → 0.133, used as the top of the tornado range.

Each candidate's utility is a product of stated weights: evidence strength × applicability to the state × a safety factor × a control factor (a route whose control failed is penalised). **This is a transparent triage heuristic, not a calibrated posterior**, and the summary file says so in its first line.

8.1 The decision

state	prior	best action	utility
missense **unstable**	0.55	**gliclazide**, tied with **migalastat**	0.70
missense **inactive**	0.45	**dasatinib**	0.40
either, readthrough feasible	**0.00**	unchanged	—

Overall best: **gliclazide, tied with migalastat, EU 0.385**. Ties are reported, never broken silently.

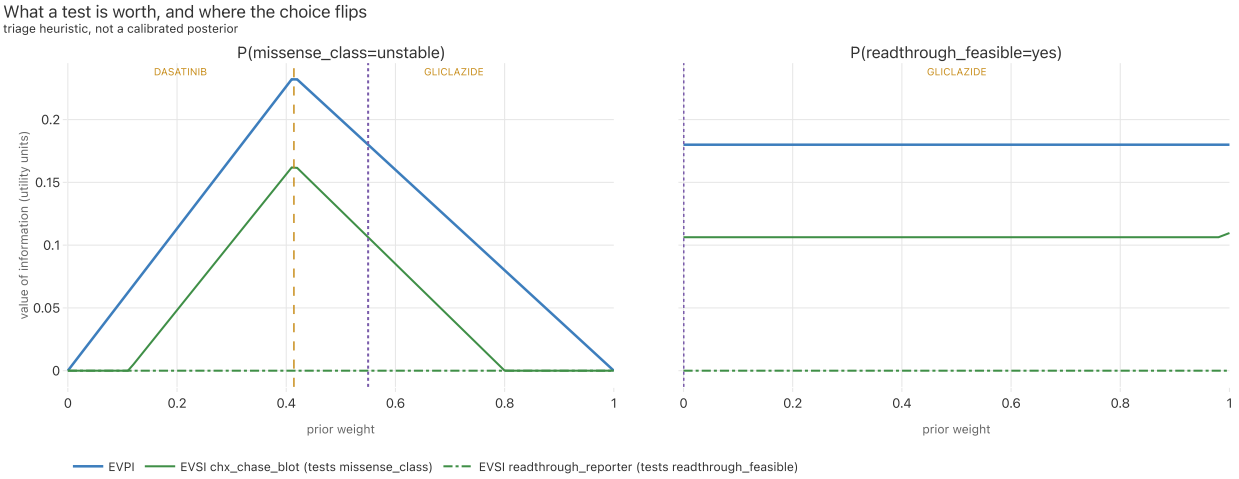


> **Legend** white → green = heuristic utility, 0 to the maximum · yellow outline = the best action in that state, with every tied action outlined · row label = the action and its per-child safety class.

8.2 The part that matters: value of information

quantity	value	reading
EVPI (perfect knowledge of everything)	**0.180**	the whole ceiling
EVPPI on `missense_class`	**0.180**	**all** of it
EVPPI on `readthrough_feasible`	**0.000**	consistent with §5
EVSI, cycloheximide-chase blot (sens/spec 0.85/0.85, a stated assumption)	**0.106**	**59 %** of the ceiling, positive across the whole tornado range (0.005–0.150)
EVSI, readthrough reporter assay	0.000	not worth running

The flip point. The best action changes at **P(unstable) = 0.414** — dasatinib below, gliclazide/migalastat above. Our prior of 0.55 sits only **0.14** above that line.

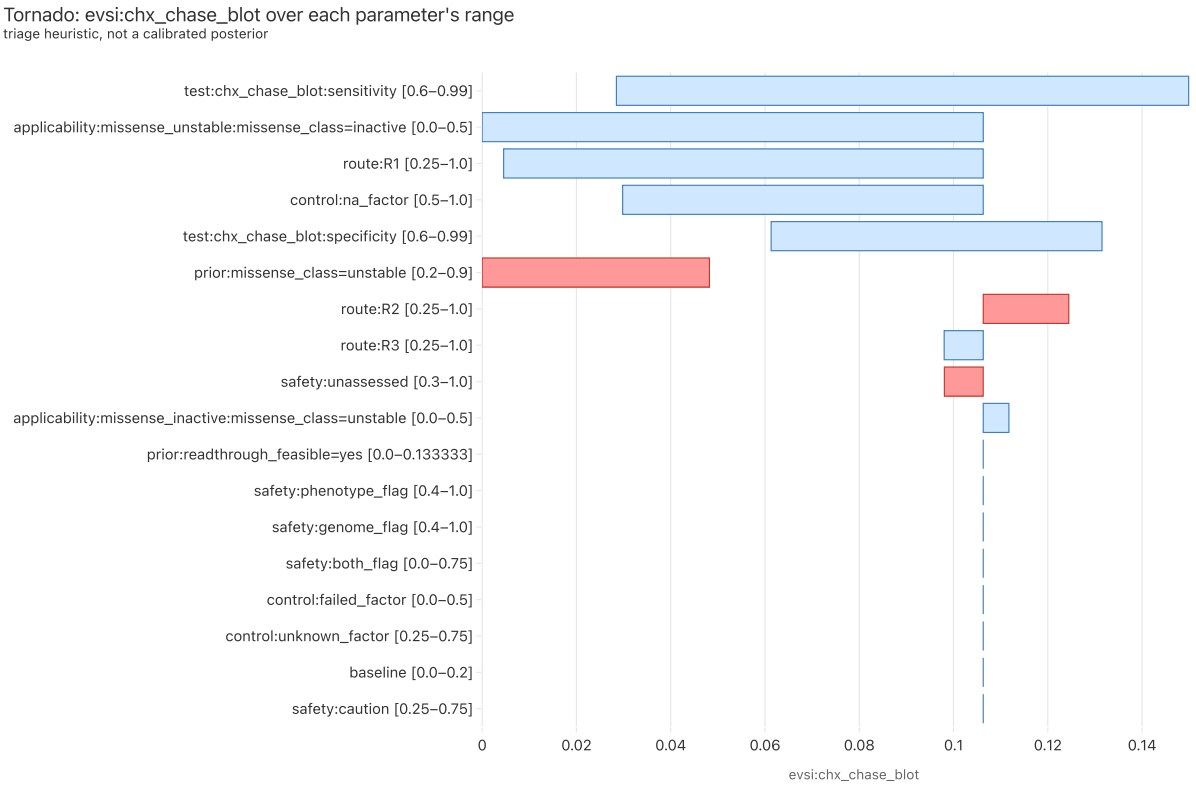


> **Legend** blue = EVPI, the upper bound · green = EVSI of a named test, one dash style per test, the legend naming the variable it measures · yellow dashed = the exact flip point, with each segment labelled by its best action · purple dotted = the configured prior (0.55).

8.3 What is *not* robust, said before anyone asks

The tornado flips the recommended drug in three places: at P(unstable) 0.2 (→ dasatinib), at R2 route weight 0.25 (→ NMN), and when unassessed drugs are given full weight (→ NMN).

So the drug at the top is not robust; the value of the blot is. The blot's EVSI stays positive over every range we swept. That asymmetry is the single most useful output of this iteration, and it is why §9 leads with a measurement rather than a molecule.



> **Legend** blue bar = the metric's range as one parameter moves low → high with the rest held · red = the best action changes somewhere within that range.

---

9. Potential impact

> **Rubric:** Impact (25 %). What a treating team or a lab could do with this, and what they should not do.

9.1 One measurement, ahead of any prescription

A cycloheximide-chase Western blot on the child's cells against his parents' — block new protein synthesis, then measure how fast BubR1 disappears. It answers *unstable vs inactive* directly, overriding every prediction in this report, and §8 prices it at **59 % of everything perfect knowledge could buy**. No drug on our list is worth anything close to that.

A second, cheaper experiment worth naming: **allele-specific expression** measurement on the child's transcripts, to confirm how much of the stop allele's message survives NMD. No such measurement exists for `BUB1B` in any MVA patient (§12), which is a gap this case could close.

9.2 The avoid list — the part we are most confident in

35 rows, each tied to its source. Unlike the candidates, these rest on converging evidence:

avoid	why	source
Mps1/TTK and Aurora inhibitors	weakening a checkpoint that is already weak; 33 of 34 are *less* effective in high-aneuploidy lines	entry 151, reproducing PMID 33505028
9 CHEK1 inhibitors, bomedemstat (KDM1A)	knockdowns of these targets **lower** BubR1 — the wrong direction	entry 160, PMID 19778378
geldanamycin / HSP90 inhibitors	same, wrong direction	entries 159, 160
gentamicin, amikacin, G418	nephrotoxic against the child's **nephrocalcinosis (HP:0000121)**, and the +4 A context makes them the weakest readthrough class for *this* stop	entries 158, 161
dactinomycin	severe reaction in a **biallelic-** `BUB1B` patient	entry 160, PMID 31081598
vincristine, paclitaxel	advised against in a `BUB1B` MVA report; vincristine separately flagged by the child's own CYP3A5 call and by muscle-atrophy label matching to his **HP:0003701**	entries 152, 155, 160
13 spindle poisons (caution, not exclusion)	they act through the checkpoint the child lacks	iteration 4 §4 (A2)
somatropin (caution)	ineffective across 4 MVA reports	entry 160

Three of these classes are also on the candidate list from other routes, and are driven to utility **0** by the avoid layer: G418, gentamicin, geldanamycin. A pipeline without a direction-aware avoid layer would have proposed all three.

9.3 Safety coverage, stated as coverage and not as safety

Entry 152's per-child method was re-run on the 47 iteration-5 names with no new downloads. Of 45 drugs: **9 unflagged, 7 phenotype-flagged, 2 genome-flagged, 24 unassessed** (no OnSIDES label), 3 avoid. The furosemide and calcitriol controls both reproduced.

"Unflagged" never means safe — it means two specific sources were silent. 24 of 45 drugs were not assessable at all, and the delivered callset answers only **1.6 %** of pharmacogene positions (entry 152).

---

10. What is new here, against standard practice

> **Rubric:** Innovation (25 %).

idea	standard practice	what iteration 5 did
therapy priced **per allele**	a gene gets a drug list	the stop codon's own context predicts readthrough, and a dosage arithmetic converts it into percentage points of working protein — which is what ruled the class out
the confound is the analysis	screen for "drugs that raise gene X"	BUB1B is 99.7th-percentile proliferation-bound, so the score *is* the residual after removing growth; the uncorrected version would have produced a longer, wronger list
a **code-enforced quote gate** on model-read literature	an LLM summarises papers	every row's quote must be a verbatim substring of the cached source, and the gate is mutation-tested
direction checks and an avoid layer	related drugs are proposed	knockdowns that lower the protein produce *avoid* rows, not candidates; three proposed drugs are driven to zero
decide, then price the next measurement	a ranked list	EVPI/EVSI: the bench test is worth 59 % of perfect knowledge, and the top-ranked drug is not robust — both stated
a **failed control ends a route**	a weak result is reported anyway	two of six controls failed; both routes are reported as uninterpretable and contribute nothing

11. Scalability: the same three routes on a second disease

> **Rubric:** Scalability (15 %). **Entry:** 162.

All three discovery routes were re-run on **Bloom syndrome (`BLM`)** — the same second disease iteration 4 used, so the costs are directly comparable.

route	new card	config lines	code ±	control	wall clock
R1 readthrough + dosage	2 files	311	+330 / –135	**passed**, plus an external check	6.1 s
R2 proteomics	**none**	**0**	+47 / –59	**passed** (dataset-level, reused)	12.5 s
R3 literature mining	1 file	276	+114 / –41	**FAILED** (extraction 0/3)	738 s
total	3	**587**	+491 / –235		~12.6 min

The yardstick: entry 156's transfer of the iteration-4 chain to the same disease needed a **378-line purpose-built driver plus a 24-rule relabelling pass**. Here it is 587 lines of *configuration* (case cards, no logic) plus **271 lines of one-time parameterisation** that every future case inherits. R2 transferred with **no card at all**.

Three things we report against ourselves.

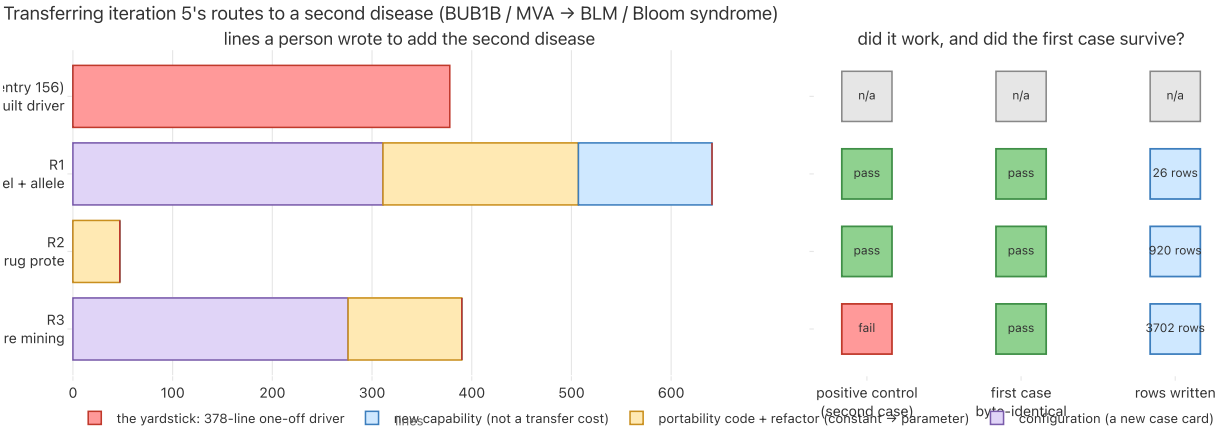
1. **"Configuration only" was false as the routes stood.** Each of the three had a constant hiding in a shared module (the effect vocabulary, the directness scale, the target gene, the case name, the manifest prose). All are now parameters, guarded by `tests/test_drugs_case_cards.py`. We report both **271** (the parameterisation) and **357** (that plus the five module splits the repo's own 400-line size gate then forced). Neither counts the 134 lines of *new capability* the transfer added, which are not a transfer cost; the full added diff is 491 lines.
2. **R3's declared control failed: retrieval 3/3, extraction 0/3.** The diagnosis is that two of the three famous Bloom papers we named in advance are genotype–phenotype *observations*, not interventions, so the screen correctly dropped them — **the control was mis-specified, not the code**. All 5 BLM candidate rows carry `positive_control_passed = false` and are reported as

uninterpretable. A corrected control is named in the entry and marked **post-hoc**, so it is not a validation.

3. **A residual case-shaped leak, found by the transfer and left standing.** `readthrough_tables.write_dosage` hard-codes the labels `stop` / `missense` as column names; for the BLM homozygote both alleles are stops, so the per-allele columns are wrong (totals are right). Fixing it would rename a column in the first case's frozen table and break the byte-identity check, so it is recorded as a known defect for the next iteration.

Every one of the nine first-case tables is byte-identical after the parameterisation and again after the module split — checked twice, digests in the entry.

A real finding came out of the transfer. Toledano 2024 measured a `BLM` premature stop directly, so its context was deleted from the fit and predicted blind: **Spearman 0.70 across 9 compounds (p 0.036), MAE 0.41 pp** — the first time any readthrough number in this iteration met a measurement of the same allele. But the model **over-predicts by 3–4× for the two compounds it ranks highest** (SJ6986, SRI: predicted ~1.2 %, measured ~0.28 %). §5's conclusion does not flip — the error runs optimistic and the number was already far too small to matter — but any future quotation of those values needs this error bar.



> **Legend** purple = configuration (a case card) · yellow = code that turned a constant into a parameter · blue = a new capability, which is not a transfer cost · green = a check that passed · red = the one-off driver this replaces.

---

12. What this report cannot tell you

> **Rubric:** Rigor. The limits, next to the results they bound.

- **No pre-registered claims.** This iteration files none (§3). Positive controls are a weaker guard against steering than pre-registration, and we say so rather than implying otherwise.
- **The top-ranked drug has no mechanism argument.** Gliclazide and migalastat rest on a 14–16 % protein rise in one colorectal cancer line. Three separate assumptions flip the top of the list.
- **Two of six controls failed** (corrected LINCS; R3 on Bloom). Those routes contribute nothing.
- **The mechanism class is predicted, not measured.** Everything in §8 turns on a variable no experiment in this repo has measured — which is exactly why §9.1 leads with the blot.
- **The readthrough model over-predicts its own top compounds by 3–4×** (§11).
- **Nothing was measured in this child's cells.** Every protein number comes from cancer cell lines or mice; no ASE or NMD-rescue measurement exists for `BUB1B` in any MVA patient.
- **A known defect is open** in the dosage table's per-allele columns for homozygous cases (§11).

- **Safety sources are silent more often than they speak**: 24 of 45 drugs unassessed; the callset answers 1.6 % of pharmacogene positions.
- **One route could not be run at all**: decryptE (MassIVE MSV000093659) was unreachable from this machine. It is stated, not substituted.

---

13. Reproducibility

> **Rubric:** Rigor. Every table in this report regenerates from committed code.

```
# R1 — readthrough context model, held-out control, allele dosage
uv run python tools/it5_readthrough.py run
uv run python tools/render_it5_r1.py

# R2 — proliferation-corrected proteomics; the corrected LINCS control
uv run python tools/it5_proteome.py run
uv run python tools/it5_lincs_cc.py run
uv run python tools/render_it5_r2.py

# R3 — corpus, quote gate, translation to approved analogues
uv run python tools/it5_litmine.py run
uv run python tools/render_it5_r3.py

# R4 — decision, VOI, avoid list
uv run python tools/it5_decision.py run \
  --candidates 'ressources/track2/it5/*_candidates.tsv' \
  --safety ressources/track2/it4/candidate_safety.tsv \
  --out ressources/track2/it5
uv run python tools/render_it5_r4.py --work ressources/track2/it5 --out figures/it5

# The second disease
uv run python tools/it5_transfer_summary.py
uv run python tools/render_it5_s1.py

# This report
uv run python tools/md_to_html.py submissions/track2/iteration-5/report.md \
  build/track2-it5-report.html --repo-url https://github.com/agharsallah/mva-hackathon/blob/ma
node tools/md_to_pdf.mjs build/track2-it5-report.html \
  submissions/track2/iteration-5/agharsallah_track2_report.pdf
```

Provenance. Every `.tsv` in `ressources/track2/it5/` has a `.provenance.json` naming its inputs with sha256, its parameters, its assumptions, its limitations and the methodology entry that produced it. Lineage edges are in `warehouse.data_lineage` under `track2.it5.*`.

Data used. Toledano 2024 (PMID 39174735); Mitchell 2023 (PMID 36593396); Saei 2019 (PMID 31844049); Nusinow 2020; Mathieson 2018; LINCS L1000 2021; ChEMBL 37; openFDA; OnSIDES v3.1.1; Europe PMC; Ensembl 116; UniProt O60566; MGI; PharmCAT.

---

14. Where every claim lives

what	where
the stop allele's anatomy, the readthrough control, the dosage arithmetic	entry 158
proliferation correction, the HSPA1A control, the corrected LINCS failure	entry 159
the corpus, the quote gate, the direction check, MVA exposures	entry 160
the decision, EVPI/EVSI, the flip points, the avoid list	entry 161
the second disease and what the transfer cost	entry 162
why no hypotheses were filed	the note
the step's written summary and one caption per figure	`narrative.json`
every term used here, defined for a beginner	`docs/dictionary.md`, chapters 30 (the routes) and 31 (deciding, and proving it transfers)

Appendix A — every drug any route named

The full candidate set, with route, branch, evidence kind, strength, whether its route's control passed, and its safety class, is `ressources/track2/it5/decision_utilities.tsv`; the avoid rows with their sources are `decision_avoid.tsv`. Top of the ranking:

rank	drug	routes	EU
1	**gliclazide** (tied)	R2	0.385
1	**migalastat** (tied)	R2	0.385
3	NMN	R1, R3	0.264
4	dasatinib	R1, R3	0.254
5	ethoxzolamide · nisoldipine · testolactone (tied)	R2	0.231
8	MG132 *(not approved)*	R1, R3	0.165
9	quercetin	R1, R3	0.130
10	niacin	R3	0.106

Read this table with §8.3 and §12 in hand: the ordering is a triage heuristic, three assumptions flip its top, and the report's actual recommendation is the measurement in §9.1.